WATER POLLUTION AND RIVER CLEANING MISSIONS

SOURCE-PATHWAY MAP → STANDARD FIREWALL → ORGANIC-POLLUTION DIAGNOSTIC → EUTROPHICATION CHAIN → INSTITUTION AND APPROVAL LADDER → SEWAGE QUANTITY LEDGER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW · ● SUBORDINATE EXTRA · ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 SOURCE-PATHWAY MAP

SOURCE-PATHWAY MAP · POINT SOURCE · IDENTIFIABLE PIPE, DRAIN OR OUTFALL · NON-POINT SOURCE · DIFFUSE CATCHMENT RUNOFF · SOURCE TO RECEIVING WATER · SOURCE SAMPLE VERSUS CATCHMENT EVIDENCE · TOOL MUST MATCH THE PATHWAY

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
POINT SOURCE → identifiable pipe, drain or outfall	PATHWAY → source to receiving water	CONTROL → tool must match the pathway
NON-POINT SOURCE → diffuse catchment runoff	MONITORING → source sample versus catchment evidence	POINT SOURCE → identifiable pipe, drain or outfall

ECOLOGICAL UNIT / PARAMETER
- POINT SOURCE → identifiable pipe, drain or outfall
- NON-POINT SOURCE → diffuse catchment runoff

DECISIVE DISTINCTION
- PATHWAY → source to receiving water
- MONITORING → source sample versus catchment evidence

SCALE / STATUS / EXCEPTION
- CONTROL → tool must match the pathway
- POINT SOURCE → identifiable pipe, drain or outfall

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POINT SOURCE → identifiable pipe, drain or outfall.

01

STAGE 01 STANDARD FIREWALL

STANDARD FIREWALL · EFFLUENT STANDARD · CONTROLS DISCHARGE FROM A SOURCE · WATER-QUALITY CRITERION · DESCRIBES RECEIVING WATER · USE CLASS · INTENDED-WATER-USE FRAMEWORK · UNIT AND SAMPLING

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
EFFLUENT STANDARD → controls discharge from a source	USE CLASS → intended-water-use framework	RULE → never exchange these values
WATER-QUALITY CRITERION → describes receiving water	UNIT AND SAMPLING → belong to the exact indicator	EFFLUENT STANDARD → controls discharge from a source

ECOLOGICAL UNIT / PARAMETER
- EFFLUENT STANDARD → controls discharge from a source
- WATER-QUALITY CRITERION → describes receiving water

DECISIVE DISTINCTION
- USE CLASS → intended-water-use framework
- UNIT AND SAMPLING → belong to the exact indicator

SCALE / STATUS / EXCEPTION
- RULE → never exchange these values
- EFFLUENT STANDARD → controls discharge from a source

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UNIT AND SAMPLING → belong to the exact indicator.

02

STAGE 02 ORGANIC-POLLUTION DIAGNOSTIC

ORGANIC-POLLUTION DIAGNOSTIC · ORGANIC LOAD · MICROBIAL DECOMPOSITION · OXYGEN DEMAND DURING DECOMPOSITION · OXYGEN REMAINING FOR AQUATIC LIFE · FAECAL INDICATOR · CONTAMINATION WARNING · USE INDICATORS TOGETHER, NOT INTERCHANGEABLY

SYSTEM INPUT / STARTING CONDITION	ECOLOGICAL PROCESS / TRANSFER	OUTCOME / FEEDBACK / LIMIT
ORGANIC LOAD → microbial decomposition	DO → oxygen remaining for aquatic life	DIAGNOSIS → use indicators together, not interchangeably
BOD → oxygen demand during decomposition	FAECAL INDICATOR → contamination warning	ORGANIC LOAD → microbial decomposition

SYSTEM INPUT / STARTING CONDITION
- ORGANIC LOAD → microbial decomposition
- BOD → oxygen demand during decomposition

ECOLOGICAL PROCESS / TRANSFER
- DO → oxygen remaining for aquatic life
- FAECAL INDICATOR → contamination warning

OUTCOME / FEEDBACK / LIMIT
- DIAGNOSIS → use indicators together, not interchangeably
- ORGANIC LOAD → microbial decomposition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAECAL INDICATOR → contamination warning; DIAGNOSIS → use indicators together, not interchangeably.

03

STAGE 03 EUTROPHICATION CHAIN

EUTROPHICATION CHAIN · NUTRIENT INPUT · NITROGEN OR PHOSPHORUS ENRICHMENT · ALGAL GROWTH · BIOMASS INCREASE · OXYGEN DEMAND · LOW OXYGEN · ECOLOGICAL STRESS

NUTRIENT INPUT → nitrogen or phosphorus enrichment → ALGAL GROWTH → biomass increase → DECOMPOSITION → oxygen demand → LOW OXYGEN → ecological stress → BOUNDARY → not identical to toxic industrial discharge

SYSTEM INPUT / STARTING CONDITION
- NUTRIENT INPUT → nitrogen or phosphorus enrichment
- ALGAL GROWTH → biomass increase

ECOLOGICAL PROCESS / TRANSFER
- DECOMPOSITION → oxygen demand
- LOW OXYGEN → ecological stress

OUTCOME / FEEDBACK / LIMIT
- BOUNDARY → not identical to toxic industrial discharge
- NUTRIENT INPUT → nitrogen or phosphorus enrichment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → not identical to toxic industrial discharge.

04

STAGE 04 INSTITUTION AND APPROVAL LADDER

INSTITUTION AND APPROVAL LADDER · WATER ACT · STATUTORY POLLUTION-CONTROL FRAMEWORK · NATIONAL COORDINATION AND FRAMEWORK · SPCB OR PCC · CONSENT, MONITORING AND ENFORCEMENT · CTE OR CTO · SOURCE PERMISSION LAYER

WATER ACT → statutory pollution-control framework

CPCB → national coordination and framework → SPCB OR PCC → consent, monitoring and enforcement → CTE OR CTO → source permission layer

SYSTEM / LEVEL / DEFINITION
- WATER ACT → statutory pollution-control framework
- CPCB → national coordination and framework

STRUCTURE-FUNCTION MECHANISM
- SPCB OR PCC → consent, monitoring and enforcement
- CTE OR CTO → source permission layer

SCALE / EVIDENCE LIMIT
- ENVIRONMENTAL CLEARANCE → separate prior-appraisal layer
- WATER ACT → statutory pollution-control framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPCB OR PCC → consent, monitoring and enforcement.

05

STAGE 05 SEWAGE QUANTITY LEDGER

SEWAGE QUANTITY LEDGER · TOTAL WASTEWATER PRODUCED · FLOW REACHING A NETWORK · NAMEPLATE TREATMENT CAPACITY · ACTUALLY TREATED · OPERATING INFLOW AND PERFORMANCE · REUSED OR DISCHARGED · FINAL PATHWAY AND QUALITY

GENERATED → total wastewater produced → SEWERED → flow reaching a network → INSTALLED → nameplate treatment capacity → ACTUALLY TREATED → operating inflow and performance → REUSED OR DISCHARGED → final pathway and quality

SYSTEM / LEVEL / DEFINITION
- GENERATED → total wastewater produced
- SEWERED → flow reaching a network

STRUCTURE-FUNCTION MECHANISM
- INSTALLED → nameplate treatment capacity
- ACTUALLY TREATED → operating inflow and performance

SCALE / EVIDENCE LIMIT
- REUSED OR DISCHARGED → final pathway and quality
- GENERATED → total wastewater produced

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REUSED OR DISCHARGED → final pathway and quality.

06

STAGE 06 TREATMENT-PLANT MATRIX

TREATMENT-PLANT MATRIX · MUNICIPAL OR DOMESTIC SEWAGE · ONE ESTABLISHMENT'S INDUSTRIAL EFFLUENT · COMMON TREATMENT FOR A GROUP OF UNITS · INFRASTRUCTURE OUTPUT · COMPLIANT OPERATION · SEPARATELY VERIFIED PERFORMANCE

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
STP → municipal or domestic sewage	CETP → common treatment for a group of units	COMPLIANT OPERATION → separately verified performance
ETP → one establishment's industrial effluent	INSTALLATION → infrastructure output	STP → municipal or domestic sewage

ECOLOGICAL UNIT / PARAMETER
- STP → municipal or domestic sewage
- ETP → one establishment's industrial effluent

DECISIVE DISTINCTION
- CETP → common treatment for a group of units
- INSTALLATION → infrastructure output

SCALE / STATUS / EXCEPTION
- COMPLIANT OPERATION → separately verified performance
- STP → municipal or domestic sewage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CETP → common treatment for a group of units.

07

STAGE 07 REGULATOR-MISSION FIREWALL

REGULATOR-MISSION FIREWALL · CPCB OR SPCB · STATUTORY POLLUTION-CONTROL ROLE · RIVER MISSION · PROJECT AND BASIN COORDINATION · GANGA MISSION IMPLEMENTATION ARCHITECTURE · SEWERAGE SERVICE AND LOCAL OPERATION · ONE ACTOR NEVER SUBSTITUTES FOR ALL OTHERS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
CPCB OR SPCB → statutory pollution-control role	NMCG → Ganga mission implementation architecture	RULE → one actor never substitutes for all others
RIVER MISSION → project and basin coordination	ULB → sewerage service and local operation	CPCB OR SPCB → statutory pollution-control role

ECOLOGICAL UNIT / PARAMETER
- CPCB OR SPCB → statutory pollution-control role
- RIVER MISSION → project and basin coordination

DECISIVE DISTINCTION
- NMCG → Ganga mission implementation architecture
- ULB → sewerage service and local operation

SCALE / STATUS / EXCEPTION
- RULE → one actor never substitutes for all others
- CPCB OR SPCB → statutory pollution-control role

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → one actor never substitutes for all others.

08

STAGE 08 RIVER-MISSION COMPONENT WHEEL

RIVER-MISSION COMPONENT WHEEL · SEWAGE INFRASTRUCTURE · INTERCEPT AND TREAT FLOW · INDUSTRIAL MONITORING · SOURCE COMPLIANCE · BIODIVERSITY AND RIVERBANK MEASURES · PUBLIC PARTICIPATION · BEHAVIOUR AND ACCOUNTABILITY

SYSTEM / LEVEL / DEFINITION
- SEWAGE INFRASTRUCTURE → intercept and treat flow
- INDUSTRIAL MONITORING → source compliance

STRUCTURE-FUNCTION MECHANISM
- ECOLOGY → biodiversity and riverbank measures
- PUBLIC PARTICIPATION → behaviour and accountability

SCALE / EVIDENCE LIMIT
- VISIBLE WORK → not itself a water-quality outcome
- SEWAGE INFRASTRUCTURE → intercept and treat flow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VISIBLE WORK → not itself a water-quality outcome.

09

STAGE 09 PROGRAMME-SCOPE GATE

PROGRAMME-SCOPE GATE · GANGA QUESTION · USE NOTIFIED GANGA ARCHITECTURE · OTHER RIVER · IDENTIFY APPLICABLE PROGRAMME AND STATE ACTORS · WIDER RIVER-CONSERVATION PROGRAMME CONTEXT · RETAIN RIVER, STRETCH AND DATE · NO TRANSFER

GANGA QUESTION → use notified Ganga architecture → OTHER RIVER → identify applicable programme and state actors → NRCP → wider river-conservation programme context → RESULT → retain river, stretch and date → NO TRANSFER → never move one mission result to another river

SYSTEM / LEVEL / DEFINITION
- GANGA QUESTION → use notified Ganga architecture
- OTHER RIVER → identify applicable programme and state actors

STRUCTURE-FUNCTION MECHANISM
- NRCP → wider river-conservation programme context
- RESULT → retain river, stretch and date

SCALE / EVIDENCE LIMIT
- NO TRANSFER → never move one mission result to another river
- GANGA QUESTION → use notified Ganga architecture

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO TRANSFER → never move one mission result to another river.

10

STAGE 10 PERFORMANCE AND EVIDENCE CHAIN

PERFORMANCE AND EVIDENCE CHAIN · OUTLAY OR SANCTION · FINANCIAL ACTIVITY · ASSET COMPLETED · PHYSICAL OUTPUT · FLOW TREATED OR LOAD REDUCED · OPERATIONAL RESULT · RIVER QUALITY

OUTLAY OR SANCTION → input → EXPENDITURE → financial activity → ASSET COMPLETED → physical output → FLOW TREATED OR LOAD REDUCED → operational result → RIVER QUALITY → stretch-season-indicator outcome

SYSTEM INPUT / STARTING CONDITION
- OUTLAY OR SANCTION → input
- EXPENDITURE → financial activity

ECOLOGICAL PROCESS / TRANSFER
- ASSET COMPLETED → physical output
- FLOW TREATED OR LOAD REDUCED → operational result

OUTCOME / FEEDBACK / LIMIT
- RIVER QUALITY → stretch-season-indicator outcome
- OUTLAY OR SANCTION → input

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOW TREATED OR LOAD REDUCED → operational result.

11

STAGE 11 WATER ANSWER SPINE

WATER ANSWER SPINE · SOURCE, PATHWAY AND POLLUTANT · INDICATOR, UNIT, LOCATION, SEASON AND SAMPLING · CONSENT, EFFLUENT CONTROL AND ENFORCEMENT · SEWER NETWORK, STP, ETP OR CETP OPERATION · LOAD AND RIVER OUTCOME, WITH PYQ AND LIVE-DATA LIMITS

DIAGNOSE → source, pathway and pollutant → MEASURE → indicator, unit, location, season and sampling → REGULATE → consent, effluent control and enforcement → TREAT → sewer network, STP, ETP or CETP operation → JUDGE → load and river outcome, with PYQ and live-data limits

SYSTEM / LEVEL / DEFINITION
- DIAGNOSE → source, pathway and pollutant
- MEASURE → indicator, unit, location, season and sampling

STRUCTURE-FUNCTION MECHANISM
- REGULATE → consent, effluent control and enforcement
- TREAT → sewer network, STP, ETP or CETP operation

SCALE / EVIDENCE LIMIT
- JUDGE → load and river outcome, with PYQ and live-data limits
- DIAGNOSE → source, pathway and pollutant

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JUDGE → load and river outcome, with PYQ and live-data limits.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · RIVER WATER-QUALITY MONITORING (BOD, DO, FAECAL COLIFORM TRENDS AT · ANALYTICAL CAUTION · A CLAIM THAT "THE GANGA IS NOW CLEAN" OR · PARLIAMENTARY STANDING COMMITTEE AND CAG-TYPE REVIEWS OF LARGE RIVER-MISSION · FEDERAL COORDINATION CHALLENGE · BECAUSE THE GANGA FLOWS THROUGH MULTIPLE STATES · PRECISE, REAL-TIME FIGURES FOR SEWAGE-TREATMENT CAPACITY CREATED VERSUS ACTUALLY

OPTIONAL ECOLOGICAL PROCESS DEPTH
- CAUTION: River water-quality monitoring (BOD, DO, faecal coliform trends at various monitoring
- CAUTION: Analytical caution: a claim that "the Ganga is now clean" or "is now fit for
- CAUTION: Parliamentary standing committee and CAG-type reviews of large river-mission spending
- CAUTION: Federal coordination challenge: because the Ganga flows through multiple states

SCALE, PARAMETER OR STATUS LIMIT
- CAUTION: Precise, real-time figures for sewage-treatment capacity created versus actually
- CAUTION: Non-point-source agricultural-runoff pollution (fertiliser/pesticide-driven nutrient
- Namami Gange addresses multiple components — sewage-treatment infrastructure,
- A documented implementation gap in river-cleaning missions is the "last-mile" problem:

COMPARATIVE RESTORATION NUANCE
- River water-quality improvement is stretch-specific and season-variable; blanket
- Non-point-source agricultural-runoff pollution is measured and regulated with
- NOT: Namami Gange is solely a sewage-treatment-infrastructure programme. - It integrates
- NOT: Created sewage-treatment-plant capacity automatically means proportionate pollution-load

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels